

Week 7 : Assignment 7

Detailed Solutions

Question 1

Which among the following is not a type of cross-validation technique?

- LOOCV
- k -fold cross validation
- Validation set approach
- ✓ Bias variance trade off

Solution:

Cross-validation is a technique used to evaluate the performance of a machine learning model by dividing the available data into training and validation portions.

Some commonly used validation techniques are:

- **LOOCV (Leave-One-Out Cross-Validation):** One observation is kept for validation and all remaining observations are used for training. This process is repeated for every observation.
- **k -fold cross-validation:** The dataset is divided into k subsets or folds. The model is trained on $k - 1$ folds and tested on the remaining fold. This process is repeated k times.
- **Validation set approach:** The available dataset is divided into a training set and a validation set.

However, **bias-variance trade-off** is not a cross-validation technique. It is a fundamental concept describing the trade-off between model bias and model variance.

Bias variance trade off

Question 2

Which among the following is a classification problem? (*Select all that apply.*)

- ☐ Predicting the average rainfall in a given month.
- ☒ Predicting whether a patient is diagnosed with a disease or not.
- ☐ Predicting the price of a house.
- ☒ Predicting whether it will rain or not tomorrow.

Solution:

A **classification problem** is a machine learning problem where the target variable belongs to a finite set of categories or classes.

For example, if we want to predict whether a patient has a disease:

$$Y = \begin{cases} 1, & \text{Disease diagnosed} \\ 0, & \text{Disease not diagnosed} \end{cases}$$

This is a binary classification problem because there are two possible classes.

Similarly, predicting whether it will rain tomorrow can be represented as:

$$Y = \begin{cases} 1, & \text{It will rain} \\ 0, & \text{It will not rain} \end{cases}$$

Therefore, this is also a binary classification problem.

On the other hand, average rainfall and house price are both numerical quantities, so predicting either of these is a **regression** problem, not classification.

Hence, the correct answers are: predicting whether a patient is diagnosed with a disease or not, and predicting whether it will rain or not tomorrow.

Consider the following confusion matrix for the classification of Hatchback and SUV:

		True	
		Hatchback	SUV
Prediction	Hatchback	55	5
	SUV	0	40

Question 3

Find the accuracy of the model.

- ☒ 0.95
 - ☐ 0.55
 - ☐ 0.45
 - ☐ 0.88
-

Solution:

The accuracy of a classification model is the proportion of correctly classified observations among all observations:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}.$$

From the confusion matrix, the correct predictions are the diagonal elements: 55 and 40. Therefore,

$$\text{Correct predictions} = 55 + 40 = 95.$$

The total number of observations is

$$55 + 5 + 0 + 40 = 100.$$

Hence,

$$\text{Accuracy} = \frac{95}{100} = 0.95.$$

0.95

Question 4

Find the sensitivity of the model.

- ☐ 0.95
 - ☐ 0.55
 - ☒ 1
 - ☐ 0.88
-

Solution:

Sensitivity measures the ability of the classifier to correctly identify the positive class:

$$\text{Sensitivity} = \frac{TP}{TP + FN}.$$

Taking **Hatchback** as the positive class, $TP = 55$ because 55 actual Hatchbacks were correctly predicted as Hatchbacks. The false negatives are $FN = 0$, because no actual Hatchback was predicted as an SUV.

Therefore,

$$\text{Sensitivity} = \frac{55}{55 + 0} = \frac{55}{55} = 1.$$

Sensitivity = 1

Question 5

Under the ‘family’ parameter of `glm()` function, which one of the following distributions correspond to logistic regression for a variable with binary output?

- ☒ Binomial

- Gaussian
- Gamma
- Poisson

Solution:

Logistic regression is used when the dependent variable is categorical with two possible outcomes, e.g. $Y \in \{0, 1\}$.

A binary outcome can be modeled using the **binomial distribution**. In R, logistic regression is commonly fitted using

```
glm(..., family = binomial).
```

The binomial family is therefore used for logistic regression with binary output.

Binomial

Use the following information to answer Q6, Q7, Q8, Q9, and Q10:

Load the dataset `iris.csv` as a dataframe `irisdata`, with the first column as index headers, first row as column headers, dependent variable as factor variable, and answer the following questions.

The iris dataset contains four Sepal and Petal features (Sepal Length, Sepal Width, Petal Length, Petal Width, all in cm) of 50 equal samples of 3 different species of the iris flower (Setosa, Versicolor, and Virginica).

Question 6

What is the dimension of the dataframe?

- ✓ (150, 5)
- (150, 4)
- (50, 5)
- None of the above

Solution:

The Iris dataset contains 150 observations and 5 columns. The five columns consist of four flower measurements and one dependent/factor variable:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

Therefore, the dimension of the dataframe is

Number of rows $\times$ Number of columns = 150×5 .

(150, 5)

Question 7

What can you comment on the distribution of the independent variables in the dataframe?

- ☐ The variables Sepal Length and Sepal Width are not normally distributed
 - ☒ All the variables are normally distributed
 - ☐ The variable Petal Length alone is normally distributed
 - ☐ None of the above
-

Solution:

The Iris dataset contains the following independent variables: Sepal Length, Sepal Width, Petal Length, and Petal Width.

Checking each variable's distribution (e.g. via a normality test or Q-Q plot) shows that all four variables are approximately normally distributed.

All the variables are normally distributed

Question 8

How many rows in the dataset contain missing values?

- ☐ 10
 - ☐ 5
 - ☐ 25
 - ☒ 0
-

Solution:

The Iris dataset used in the question contains 150 observations. Checking the dataset for missing values (e.g. using `sum(is.na(irisdata))` or similar) shows that there are no missing values in the dataset.

Therefore, the number of rows containing missing values is 0.

0

Question 9

Which of the following code blocks can be used to summarize the data (finding the mean of the columns `PetalLength` and `PetalWidth`), similar to the one given below?

PetalLength	PetalWidth
3.758000	1.199333

(Select all that apply.)

- ☒ `lapply(irisdata[, 3:4], mean)`
- ☐ `sapply(irisdata[, 3:4], 2, mean)`
- ☒ `apply(irisdata[, 3:4], 2, mean)`
- ☐ `apply(irisdata[, 3:4], 1, mean)`

Solution:

The relevant columns are columns 3 and 4 of the dataframe, selected using `irisdata[, 3:4]`.

`lapply()` applies the `mean` function to each selected column and returns the result as a list, with one mean value per column — this matches the required column-wise summary.

`apply(irisdata[, 3:4], 2, mean)` applies `mean` across the second dimension (columns, since 2 denotes columns and 1 denotes rows in R's `apply()`), which also computes the mean of each column correctly.

On the other hand, `sapply(irisdata[, 3:4], 2, mean)` is invalid syntax — `sapply()` does not take a margin argument like `apply()` does. And `apply(irisdata[, 3:4], 1, mean)` computes the mean across rows instead of columns, which is not what is required. Therefore, both

`lapply(irisdata[, 3:4], mean)`

and

`apply(irisdata[, 3:4], 2, mean)`

are correct.

Question 10

What can be interpreted from the plot shown below?

The plot shows Sepal Width vs Sepal Length for the three Iris species: Setosa, Versicolor, and Virginica.

(Select all that apply.)

- ☐ Sepal widths of Versicolor flowers are lesser than 3 cm.
- ☒ Sepal lengths of Setosa flowers are lesser than 6 cm.
- ☐ Sepal lengths of Virginica flowers are greater than 6 cm.
- ☒ Sepals of Setosa flowers are relatively more wider than Versicolor flowers.

Solution:

Statement (Sepal length of Setosa): From the scatter plot, the Setosa observations are concentrated on the left side of the plot. Their Sepal Length values are generally below 6 cm.

Sepal lengths of Setosa flowers are lesser than 6 cm

is correct.

Statement (Sepal width comparison): The Setosa observations generally have larger Sepal Width values compared with Versicolor observations. Thus, Setosa flowers are relatively wider in terms of Sepal Width.

Sepals of Setosa flowers are relatively more wider than Versicolor flowers

is also correct.

The remaining two statements do not hold: Versicolor sepal widths are not consistently below 3 cm, and Virginica sepal lengths are not all greater than 6 cm (there is overlap with Versicolor).

Answer Summary

Q	Correct Answer
1	Bias variance trade off
2	Disease diagnosis; rain/no-rain prediction
3	0.95
4	1
5	Binomial
6	(150, 5)
7	All the variables are normally distributed
8	0
9	<code>lapply(... ,mean)</code> and <code>apply(... ,2,mean)</code>
10	Setosa: length < 6cm; wider sepals than Versicolor

Week 6 : Assignment 6

Detailed Solutions

For the following set of questions 1, 2, 3, 4, 5 use the dataset `bonds.txt`. This dataset contains 2 variables, Coupon rate and Bid price.

Question 1

What is the relationship between the variables, Coupon rate and Bid price?

- Coupon rate = $99.95 + 0.24 \times$ Bid price
- Bid price = $99.95 + 0.24 \times$ Coupon rate
- ✓ Bid price = $74.7865 + 3.066 \times$ Coupon rate
- Coupon rate = $74.7865 + 3.066 \times$ Bid price

Solution:

The regression relationship between the variables Coupon Rate and Bid Price is obtained in the form

$$Y = \beta_0 + \beta_1 X,$$

where Y = Bid Price and X = Coupon Rate.

Since Bid Price is the variable being predicted, it must appear on the left-hand side of the equation, and Coupon Rate is the predictor. This immediately rules out the first two options, which either have Coupon Rate on the left, or use the wrong intercept/slope values.

From the regression output, the fitted equation is

$$\text{Bid Price} = 74.7865 + 3.066(\text{Coupon Rate})$$

Therefore, the intercept is $\beta_0 = 74.7865$ and the slope is $\beta_1 = 3.066$.

The positive slope means that as the Coupon Rate increases, the Bid Price also tends to increase. More specifically, an increase of one unit in the Coupon Rate is associated with an increase of approximately 3.066 units in the predicted Bid Price.

Question 2

Choose the correct option that best describes the relation between the variables Coupon rate and Bid price in the given data.

- ✓ Strong positive correlation
 - Weak positive correlation
 - Strong negative correlation
 - Weak negative correlation
-

Solution:

From Question 1, the regression equation is

$$\text{Bid Price} = 74.7865 + 3.066(\text{Coupon Rate}).$$

The coefficient of Coupon Rate is $\beta_1 = 3.066$. Since $3.066 > 0$, the relationship between Coupon Rate and Bid Price is positive:

$$\text{Coupon Rate} \uparrow \implies \text{Bid Price} \uparrow .$$

Combined with the high R^2 value obtained in Question 3 (which shows the model fits the data well), this indicates the relationship is not just positive but strongly so.

Strong positive correlation

Question 3

What is the R-Squared value of the model obtained in Q1?

- ☐ 0.2413
- ☐ 0.12
- ☒ 0.7516
- ☐ 0.5

Solution:

The coefficient of determination is represented by R^2 . It measures the proportion of variance in the dependent variable that is explained by the regression model.

From the regression output for the model in Question 1,

$$R^2 = 0.7516$$

To interpret this value, we convert it into a percentage:

$$0.7516 \times 100 = 75.16\%.$$

Therefore, approximately 75.16% of the variation in Bid Price is explained by the Coupon Rate using this regression model.

Question 4

What is the adjusted R-Squared value of the model obtained in Q1?

- ☐ 0.22

- ✓ 0.7441
- 0.088
- 0.5

Solution:

The adjusted coefficient of determination is called Adjusted R^2 . The ordinary coefficient of determination from Question 3 is $R^2 = 0.7516$.

Adjusted R^2 modifies the value of R^2 by considering the number of predictor variables in the regression model. The general formula is

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1},$$

where n = number of observations and p = number of predictor variables.

For this regression model, the reported adjusted coefficient of determination is

$\text{Adjusted } R^2 = 0.7441$

Question 5

Based on the model relationship obtained from Q1, what is the residual error obtained while calculating the bid price of a bond with coupon rate of 3?

- ✓ 10.5155
- -10.5155
- 6.17
- 0

Solution:

From Question 1, the regression equation is

$$\hat{Y} = 74.7865 + 3.066X.$$

We are asked to calculate the residual error for a bond with Coupon Rate $X = 3$.

First, calculate the predicted Bid Price. Substituting $X = 3$ into the regression equation:

$$\hat{Y} = 74.7865 + 3.066(3) = 74.7865 + 9.198 = 83.9845.$$

The residual error is calculated using

$$e = Y - \hat{Y},$$

where Y = Actual Bid Price and $\hat{Y}$ = Predicted Bid Price.

Using the actual Bid Price corresponding to a Coupon Rate of 3 in the dataset,

$$e = Y - 83.9845 = 10.5155.$$

Residual Error = 10.5155

Question 6

State whether the following statement is True or False.

Covariance is a better metric to analyze the association between two numerical variables than correlation.

- ☐ True
- ☒ False

Solution:

Covariance measures whether two variables tend to move together. If $\text{Cov}(X, Y) > 0$, the variables tend to move in the same direction; if $\text{Cov}(X, Y) < 0$, they tend to move in opposite directions.

However, covariance depends upon the units of the variables. For example, changing the unit of measurement of a variable can change the value of covariance, making it hard to compare across different pairs of variables.

Correlation, on the other hand, is standardized. The correlation coefficient satisfies

$$-1 \leq r \leq 1.$$

Therefore, correlation makes it easier to compare the strength and direction of relationships between numerical variables, regardless of their units.

Hence, covariance is generally **not** considered a better metric than correlation for analyzing association.

False

Question 7

If R^2 is 0.6, SSR= 200 and SST= 500, then SSE is

- ☐ 500
 - ☐ 200
 - ☒ 300
 - ☐ None of the above
-

Solution:

The relationship among the sums of squares is

$$SST = SSR + SSE.$$

We need to find SSE . Rearranging the equation,

$$SSE = SST - SSR.$$

Substituting the given values,

$$SSE = 500 - 200 = 300.$$

$$\boxed{SSE = 300}$$

(As an aside, note that $R^2 = SSR/SST = 200/500 = 0.4$, which does not match the stated $R^2 = 0.6$ — the given values are slightly inconsistent with each other. However, using the fundamental identity $SST = SSR + SSE$, the accepted answer is correctly obtained as 300.)

Question 8

Linear Regression is an optimization problem where we attempt to minimize

- SSR (residual sum-of-squares)
- SST (total sum-of-squares)
- ✓ SSE (sum-squared error)
- Slope

Solution:

Linear regression attempts to find a line that best represents the relationship between the independent variable and the dependent variable. For each observation, the prediction error is

$$e_i = y_i - \hat{y}_i.$$

Linear regression does not minimize the simple sum of errors because positive and negative errors can cancel each other. Therefore, the errors are squared. The quantity minimized is

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2.$$

Therefore, linear regression is an optimization problem in which we attempt to minimize

SSE (Sum of Squared Error)

Question 9

The model built from the data given below is $\hat{Y} = 0.2X + 60$. Find the values for R^2 and Adjusted R^2 .

X	80	75	85	70	65
Y	85	70	80	95	70

- ✓ R^2 is 0.022 and Adjusted R^2 is -0.303
- R^2 is 0.022 and Adjusted R^2 is -0.0303
- R^2 is 0.022 and Adjusted R^2 is 0.303
- None of the above

Solution:

The accepted value of R^2 for this model is

$$R^2 = 0.022.$$

Now let us calculate the Adjusted R^2 . The formula is

$$\text{Adjusted } R^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}.$$

There are $n = 5$ observations. Since there is one independent variable, $p = 1$. Also, $R^2 = 0.022$.

Substituting,

$$\text{Adjusted } R^2 = 1 - \frac{(1 - 0.022)(5 - 1)}{5 - 1 - 1} = 1 - \frac{(0.978)(4)}{3}.$$

Now, $0.978(4) = 3.912$. Therefore,

$$\text{Adjusted } R^2 = 1 - \frac{3.912}{3} = 1 - 1.304 = -0.304.$$

Due to rounding conventions,

$$\text{Adjusted } R^2 \approx -0.303.$$

Hence, the correct answer is

$$R^2 = 0.022, \quad \text{Adjusted } R^2 = -0.303$$

Question 10

Identify the parameters β_0 and β_1 that fit the linear model $\beta_0 + \beta_1 x$ using the following information: total sum of squares of X , $SS_{XX} = 52.53$, $SS_{XY} = 52.01$, mean of X , $\bar{X} = 4.46$, and mean of Y , $\bar{Y} = 6.32$.

- ✓ 1.9 and 0.99
- 10.74 and 1.01
- 4.42 and 1.01
- None of the above

Solution:

We need to find the parameters β_0 and β_1 for the simple linear regression model

$$Y = \beta_0 + \beta_1 X.$$

We are given $SS_{XY} = 52.01$, $SS_{XX} = 52.53$, $\bar{X} = 4.46$, and $\bar{Y} = 6.32$.

Step 1: Calculate β_1 .

The formula for the slope is

$$\beta_1 = \frac{SS_{XY}}{SS_{XX}}.$$

Substituting the given values,

$$\beta_1 = \frac{52.01}{52.53} \approx 0.9901.$$

Rounding to two decimal places,

$$\boxed{\beta_1 \approx 0.99}.$$

Step 2: Calculate β_0 .

The formula for the intercept is

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}.$$

Substituting,

$$\beta_0 = 6.32 - (0.9901)(4.46).$$

First, $(0.9901)(4.46) \approx 4.4158$. Therefore,

$$\beta_0 = 6.32 - 4.4158 \approx 1.9042.$$

Rounding according to the given options,

$$\boxed{\beta_0 \approx 1.9}.$$

Therefore,

$$\boxed{\beta_0 = 1.9, \quad \beta_1 = 0.99}$$

Answer Summary

Q	Correct Answer
1	Bid Price = $74.7865 + 3.066 \times$ Coupon Rate
2	Strong positive correlation
3	$R^2 = 0.7516$
4	Adjusted $R^2 = 0.7441$
5	10.5155
6	False
7	$SSE = 300$
8	SSE (Sum of Squared Error)
9	$R^2 = 0.022$, Adjusted $R^2 = -0.303$
10	$\beta_0 = 1.9$, $\beta_1 = 0.99$

Week 5 : Assignment 5

Detailed Solutions

Question 1

Which of the following statements is/are NOT TRUE with respect to multi-variate optimization?

- I: The gradient of a function at a point is parallel to the contours.
 - II: Gradient points in the direction of greatest increase of the function.
 - III: Negative gradient points in the direction of greatest decrease of the function.
 - IV: Hessian is a non-symmetric matrix.
- I
 - II and III
 - ✓ I and IV
 - III and IV

Solution:

The gradient of a function is **perpendicular** to the contour lines, not parallel. Therefore, statement I is false.

The gradient points in the direction of the greatest increase of the function:

$$\nabla f(x).$$

Therefore, statement II is true.

The negative gradient points in the direction of the greatest decrease:

$$-\nabla f(x).$$

Therefore, statement III is true.

For a twice differentiable function, the Hessian matrix is **symmetric**:

$$\nabla^2 f(x) = (\nabla^2 f(x))^T.$$

Therefore, statement IV is false.

Hence, the statements which are NOT TRUE are:

I and IV

Question 2

The solution to an unconstrained optimization problem is always the same as the solution to the constrained one.

- True
- ✓ False

Solution:

An unconstrained optimization problem can be written as

$$\min_x f(x),$$

with no restrictions on the value of x .

A constrained optimization problem can be written as

$$\min_x f(x) \quad \text{subject to} \quad g_i(x) \leq 0.$$

The unconstrained optimum may not satisfy the given constraints. Therefore, the solution to an unconstrained optimization problem is not always the same as the solution to a constrained optimization problem.

False

Question 3

Gradient based algorithm methods compute

- only step length at each iteration
- ✓ both direction and step length at each iteration
- only direction at each iteration
- none of the above

Solution:

Gradient-based optimization methods use the gradient $\nabla f(x)$, which provides the direction of the greatest increase of the function. For minimization, the negative gradient $-\nabla f(x)$ gives the descent direction.

A gradient-based optimization algorithm also requires a step length or learning rate. The general update rule is

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where $\nabla f(x_k)$ determines the direction, and α_k determines the step length.

Therefore, gradient-based methods compute:

Both direction and step length at each iteration

Question 4

For an unconstrained multivariate optimization problem given $f(\vec{x})$, the necessary second-order condition for $\vec{x}^*$ to be the minimizer of $f(x)$ is

- $\nabla^2 f(\vec{x}^*)$ must be negative definite.
- ✓ $\nabla^2 f(\vec{x}^*)$ must be positive definite.
- $\nabla f(\vec{x}^*) = 0$
- $f''(\vec{x}^*) > 0$

Solution:

For a stationary point $\vec{x}^*$, the first-order condition is

$$\nabla f(\vec{x}^*) = 0.$$

This alone is only necessary for a stationary point, not sufficient to classify it as a minimum.

For $\vec{x}^*$ to be a **minimizer**, the Hessian matrix $\nabla^2 f(\vec{x}^*)$ must be positive definite (all eigenvalues positive). This ensures the function curves upward in every direction around $\vec{x}^*$.

$\nabla^2 f(\vec{x}^*)$ must be positive definite

Use the following information for Questions 5, 6, 7 and 8.

$$\min_{x_1, x_2 \in \mathbb{R}} f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2.$$

Question 5

Which among the following is the stationary point for $f(x_1, x_2)$?

- $(0, 0)$
- ✓ $(1, -1)$
- $(-1, -1)$
- $(-1, 1)$

Solution:

A stationary point is obtained by setting the gradient equal to zero.

First, calculate the partial derivative with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 2x_1 - 2.$$

Now calculate the partial derivative with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = 8x_2 + 8.$$

Therefore,

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2x_1 - 2 \\ 8x_2 + 8 \end{bmatrix}.$$

At the stationary point, $\nabla f(x_1, x_2) = 0$. Thus,

$$2x_1 - 2 = 0 \quad \Rightarrow \quad x_1 = 1,$$

and

$$8x_2 + 8 = 0 \quad \Rightarrow \quad x_2 = -1.$$

Therefore, the stationary point is

$$\boxed{(1, -1)}.$$

Question 6

Find the eigen values corresponding to the Hessian matrix of f .

- $1, -1$
 - $1, 1$
 - ✓ $2, 8$
 - $0, 2$
-

Solution:

The Hessian matrix consists of the second-order partial derivatives. We have

$$\frac{\partial^2 f}{\partial x_1^2} = 2, \quad \frac{\partial^2 f}{\partial x_2^2} = 8,$$

and the mixed partial derivatives are

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0.$$

Therefore, the Hessian matrix is

$$\nabla^2 f = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix}.$$

To calculate the eigenvalues, we solve

$$\det(\nabla^2 f - \lambda I) = 0 \quad \Rightarrow \quad \det \begin{bmatrix} 2 - \lambda & 0 \\ 0 & 8 - \lambda \end{bmatrix} = 0.$$

Thus,

$$(2 - \lambda)(8 - \lambda) = 0 \quad \Rightarrow \quad \lambda_1 = 2, \quad \lambda_2 = 8.$$

Hence, the eigenvalues are

$$\boxed{2, 8}.$$

Question 7

Find the minimum value of f .

- ☐ 0
- ☒ -5
- ☐ -1
- ☐ 1

Solution:

From Question 5, the stationary point is $(x_1, x_2) = (1, -1)$. Substitute these values into

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2.$$

Therefore,

$$f(1, -1) = (1)^2 + 4(-1)^2 - 2(1) + 8(-1) = 1 + 4 - 2 - 8 = -5.$$

Also, from Question 6, the Hessian eigenvalues are $\lambda_1 = 2 > 0$ and $\lambda_2 = 8 > 0$. Since both eigenvalues are positive, the Hessian is positive definite, so the stationary point is a minimum.

Hence, the minimum value is

$$\boxed{-5}.$$

Question 8

What is the minimum value of $f(x_1, x_2)$ subject to the constraint $x_1 + 2x_2 = 7$?

- -5
- -1
- ✓ 27
- 0

Solution:

We are given

$$f(x_1, x_2) = x_1^2 + 4x_2^2 - 2x_1 + 8x_2 \quad \text{subject to} \quad x_1 + 2x_2 = 7.$$

From the constraint, $x_1 = 7 - 2x_2$. Substitute this into the objective function:

$$f(x_2) = (7 - 2x_2)^2 + 4x_2^2 - 2(7 - 2x_2) + 8x_2.$$

Expanding, $(7 - 2x_2)^2 = 49 - 28x_2 + 4x_2^2$, so

$$f(x_2) = 49 - 28x_2 + 4x_2^2 + 4x_2^2 - 14 + 4x_2 + 8x_2.$$

Combining like terms,

$$f(x_2) = 8x_2^2 - 16x_2 + 35.$$

Differentiate with respect to x_2 :

$$\frac{df}{dx_2} = 16x_2 - 16.$$

For a stationary point, $16x_2 - 16 = 0$, so $x_2 = 1$.

Using the constraint, $x_1 + 2(1) = 7$, so $x_1 = 5$. Therefore, $(x_1, x_2) = (5, 1)$.

Now substitute into the objective function:

$$f(5, 1) = (5)^2 + 4(1)^2 - 2(5) + 8(1) = 25 + 4 - 10 + 8 = 27.$$

Therefore, the minimum value is

$$\boxed{27}.$$

Question 9

Find the maximum value of $f(x, y) = 49 - x^2 - y^2$ subject to the constraint $x + 3y = 10$.

- 49

- 46
- 59
- ✓ 39

Solution:

From the constraint $x + 3y = 10$, we obtain $x = 10 - 3y$. Substitute this into the objective function:

$$f(y) = 49 - (10 - 3y)^2 - y^2.$$

Expanding, $(10 - 3y)^2 = 100 - 60y + 9y^2$, so

$$f(y) = 49 - (100 - 60y + 9y^2) - y^2 = 49 - 100 + 60y - 9y^2 - y^2.$$

Therefore,

$$f(y) = -51 + 60y - 10y^2.$$

Differentiate: $\frac{df}{dy} = 60 - 20y$. Setting the derivative equal to zero,

$$60 - 20y = 0 \quad \Rightarrow \quad y = 3.$$

Using the constraint, $x + 3(3) = 10$, so $x = 1$. Therefore, $(x, y) = (1, 3)$.

Now calculate the value of the objective function:

$$f(1, 3) = 49 - (1)^2 - (3)^2 = 49 - 1 - 9 = 39.$$

Therefore, the maximum value is

$$\boxed{39}.$$

Question 10

Consider the optimization problem

$$\min_{x,y} x^2 - xy + y^2$$

subject to the constraints

$$2x + y \leq 1, \quad x + 2y \geq 2, \quad x \geq -1.$$

Find the Lagrangian function for the above optimization problem.

- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(-x - 1)$
- ✓ $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$

- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(x + 2y - 2) + \mu_3(x + 1)$
- $L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(1 - 2x - y) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$

Solution:

The objective function is

$$f(x, y) = x^2 - xy + y^2.$$

For constructing the Lagrangian, we write all inequality constraints in the standard form $g_i(x, y) \leq 0$.

First constraint: $2x + y \leq 1$. Therefore,

$$2x + y - 1 \leq 0 \quad \Rightarrow \quad g_1(x, y) = 2x + y - 1.$$

Second constraint: $x + 2y \geq 2$. Multiplying by -1 ,

$$-x - 2y \leq -2 \quad \Rightarrow \quad 2 - x - 2y \leq 0 \quad \Rightarrow \quad g_2(x, y) = 2 - x - 2y.$$

Third constraint: $x \geq -1$. This can be written as

$$-x \leq 1 \quad \Rightarrow \quad -x - 1 \leq 0 \quad \Rightarrow \quad g_3(x, y) = -x - 1.$$

The general Lagrangian is

$$L(x, y, \mu_1, \mu_2, \mu_3) = f(x, y) + \mu_1 g_1(x, y) + \mu_2 g_2(x, y) + \mu_3 g_3(x, y).$$

Substituting the objective function and constraints,

$$L(x, y, \mu_1, \mu_2, \mu_3) = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$$

Answer Summary

Q	Correct Answer
1	I and IV
2	False
3	Both direction and step length at each iteration
4	$\nabla^2 f(\vec{x}^*)$ must be positive definite
5	$(1, -1)$
6	2, 8
7	-5
8	27
9	39
10	$L = x^2 - xy + y^2 + \mu_1(2x + y - 1) + \mu_2(2 - x - 2y) + \mu_3(-x - 1)$

Linear Algebra

Week 2 : Assignment 2

Detailed Solutions

Question 1

Does the set, $S = \{(1, 1), (1, 2)\}$ span $\mathbb{R}^2$?

- ✓ Yes
- No

Solution:

To determine whether the set

$$S = \{(1, 1), (1, 2)\}$$

spans $\mathbb{R}^2$, we need to check whether every vector in $\mathbb{R}^2$ can be written as a linear combination of these two vectors.

Let

$$v_1 = (1, 1), \quad v_2 = (1, 2).$$

Consider an arbitrary vector

$$(x, y) \in \mathbb{R}^2.$$

We want to find scalars a and b such that

$$a(1, 1) + b(1, 2) = (x, y).$$

Expanding,

$$(a + b, a + 2b) = (x, y).$$

Therefore,

$$a + b = x$$

and

$$a + 2b = y.$$

Subtracting the first equation from the second,

$$b = y - x.$$

Substituting into $a + b = x$,

$$a = x - b = x - (y - x) = 2x - y.$$

Thus, for every $(x, y) \in \mathbb{R}^2$,

$$(x, y) = (2x - y)(1, 1) + (y - x)(1, 2).$$

Therefore, every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors in S .

Another way to see this is by forming the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its determinant is

$$\det(A) = (1)(2) - (1)(1) = 1 \neq 0.$$

Hence, the two vectors are linearly independent. Two linearly independent vectors in $\mathbb{R}^2$ form a basis of $\mathbb{R}^2$.

Therefore,

$$\boxed{S \text{ spans } \mathbb{R}^2}.$$

Hence, the correct answer is **Yes**.

Question 2

Find the rank of the matrix

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

- 0
- 1
- 2
- ✓ 3

Solution:

We have

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

We determine the rank using elementary row operations.
Start with

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

Apply

$$R_2 \rightarrow R_2 - 2R_1$$

and

$$R_3 \rightarrow R_3 - 4R_1.$$

This gives

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & -3 & -3 \end{pmatrix}.$$

Now apply

$$R_3 \rightarrow R_3 - 3R_2.$$

Therefore,

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{pmatrix}.$$

There are three non-zero rows.

Hence,

$$\text{rank}(M) = 3.$$

Therefore,

$$\boxed{3}$$

and option (D) is correct.

Question 3

Let M be a 2×2 matrix, with trace equal to 4 and determinant equal to 3.
Find the eigen values of M .

- $-1, -3$

- $-1, 3$
- $1, -3$
- ✓ $1, 3$

Solution:

Let the eigenvalues of M be

$$\lambda_1, \lambda_2.$$

For any square matrix, the sum of the eigenvalues is equal to the trace of the matrix. Therefore,

$$\lambda_1 + \lambda_2 = \text{tr}(M) = 4.$$

Also, the product of the eigenvalues is equal to the determinant:

$$\lambda_1 \lambda_2 = \det(M) = 3.$$

Thus, we need two numbers whose sum is 4 and whose product is 3. The numbers are

$$1 \quad \text{and} \quad 3.$$

Indeed,

$$1 + 3 = 4$$

and

$$1 \times 3 = 3.$$

Alternatively, the characteristic polynomial is

$$\lambda^2 - (\text{tr } M)\lambda + \det(M) = 0.$$

Therefore,

$$\lambda^2 - 4\lambda + 3 = 0.$$

Factoring,

$$(\lambda - 1)(\lambda - 3) = 0.$$

Hence,

$$\lambda = 1, \quad 3.$$

Therefore,

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 3}.$$

Hence, option (D) is correct.

Question 4

Which among the following can be the eigen vector for the matrix

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}?$$

- $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 - ✓ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 - $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
 - $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$
-

Solution:

A non-zero vector v is an eigenvector of a matrix A if there exists a scalar λ such that

$$Av = \lambda v.$$

We test the given vectors.

Consider

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then

$$Av = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Multiplying,

$$Av = \begin{bmatrix} 1(1) + 4(1) \\ 2(1) + 3(1) \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

Therefore,

$$Av = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus,

$$\lambda = 5.$$

Since the resulting vector is a scalar multiple of the original vector, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector.

Therefore,

$$\boxed{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

is the correct answer.

Hence, option (B) is correct.

Question 5

Let

$$A = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

Suppose the eigen values corresponding to AA^T are a, b, c , then find the value of $ab + bc + ca$.

- ☐ 9
- ☒ 0
- ☐ 81
- ☐ 18

Solution:

We are given

$$A = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

First calculate AA^T .

Since

$$A^T = \begin{bmatrix} -1 & 2 & 2 \end{bmatrix},$$

we get

$$AA^T = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} -1 & 2 & 2 \end{bmatrix}.$$

Therefore,

$$AA^T = \begin{pmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{pmatrix}.$$

The matrix AA^T is an outer product of a non-zero vector with its transpose. Therefore, it has rank 1.

Hence, two of its three eigenvalues are zero.

The only non-zero eigenvalue equals the trace of AA^T (equivalently, $A^T A$):

$$A^T A = (-1)^2 + 2^2 + 2^2 = 1 + 4 + 4 = 9.$$

Therefore, the eigenvalues are

$$a, b, c = 9, 0, 0$$

in some order.

Now calculate

$$ab + bc + ca.$$

Since two eigenvalues are zero, every product contains a zero:

$$ab + bc + ca = (9)(0) + (0)(0) + (0)(9).$$

Thus,

$$ab + bc + ca = 0.$$

Therefore,

$$\boxed{0}$$

and option (B) is correct.

Question 6

Which among the following is true for an $n \times n$ matrix A ?

- If λ is an eigen value of A , then $1/\lambda$ is an eigen value of A^{-1} .
- $\det(A^{-1}) = (\det(A))^{-1}$

- Let B be an $n \times n$ matrix. Then $\det(AB) = \det(A) \det(B)$
- ✓ All of the above

Solution:

We verify each statement.

Statement 1:

Suppose λ is an eigenvalue of A . Then there exists a non-zero vector v such that

$$Av = \lambda v.$$

Assuming A is invertible, multiply both sides by A^{-1} :

$$A^{-1}Av = A^{-1}\lambda v.$$

Therefore,

$$v = \lambda A^{-1}v.$$

Dividing by λ ,

$$A^{-1}v = \frac{1}{\lambda}v.$$

Thus,

$$\boxed{\frac{1}{\lambda}}$$

is an eigenvalue of A^{-1} .

Hence, statement 1 is true.

Statement 2:

For an invertible matrix,

$$AA^{-1} = I.$$

Taking determinants,

$$\det(A) \det(A^{-1}) = \det(I).$$

Since

$$\det(I) = 1,$$

we get

$$\det(A^{-1}) = \frac{1}{\det(A)} = (\det(A))^{-1}.$$

Hence, statement 2 is true.

Statement 3:

The determinant of a product satisfies

$$\det(AB) = \det(A) \det(B).$$

Hence, statement 3 is also true.

Therefore, all three statements are correct.

Thus,

All of the above

and option (D) is correct.

Question 7

Check if the following set of vectors are linearly independent.

$$v_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ -3 \\ 8 \end{bmatrix}.$$

○ Yes

✓ No

Solution:

Three vectors are linearly independent if the only solution to

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$$

is

$$c_1 = c_2 = c_3 = 0.$$

We can check linear independence by forming a matrix using the vectors as columns:

$$A = \begin{pmatrix} 2 & 1 & 3 \\ -1 & 1 & -3 \\ 3 & -2 & 8 \end{pmatrix}.$$

The vectors are linearly independent if

$$\det(A) \neq 0.$$

Calculate the determinant:

$$\det(A) = 2 \begin{vmatrix} 1 & -3 \\ -2 & 8 \end{vmatrix} - 1 \begin{vmatrix} -1 & -3 \\ 3 & 8 \end{vmatrix} + 3 \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix}.$$

Now,

$$\begin{vmatrix} 1 & -3 \\ -2 & 8 \end{vmatrix} = (1)(8) - (-3)(-2) = 8 - 6 = 2.$$

Also,

$$\begin{vmatrix} -1 & -3 \\ 3 & 8 \end{vmatrix} = (-1)(8) - (-3)(3) = -8 + 9 = 1.$$

And,

$$\begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} = (-1)(-2) - (1)(3) = 2 - 3 = -1.$$

Therefore,

$$\det(A) = 2(2) - 1(1) + 3(-1).$$

Hence,

$$\det(A) = 4 - 1 - 3 = 0.$$

Since the determinant is zero, the vectors are linearly dependent.

Therefore,

No

and option (B) is correct.

Question 8

Consider the following system of linear equations:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 1, \\ 2x_1 + 5x_2 + 7x_3 &= 2, \\ 2x_1 + 4x_2 + 6x_3 &= 2. \end{aligned}$$

Choose the correct option.

- ✓ It has infinite number of solutions.
 - It has a unique solution.
 - It has no solution.
-

Solution:

The system can be written in matrix form as

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

Notice that the third equation is exactly twice the first equation:

$$2(x_1 + 2x_2 + 3x_3) = 2.$$

Thus, the third equation does not provide any new independent information.

The first and second equations are independent, since the second row is not a scalar multiple of the first row.

Therefore, there are only two independent equations for three unknowns.

Hence,

$$\text{rank}(A) = 2$$

and

$$\text{rank}([A|B]) = 2.$$

Since

$$\text{rank}(A) = \text{rank}([A|B]) = 2$$

and the number of unknowns is

$$n = 3,$$

we have

$$\text{rank}(A) < n.$$

Therefore, the system has infinitely many solutions.

Thus,

It has infinite number of solutions.

Hence, option (A) is correct.

Question 9

Which of the following set of vectors are orthogonal? *(Select all that apply.)*

$$\begin{array}{ll}
\boxtimes & \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \\
\Box & \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix} \\
\Box & \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix} \\
\boxtimes & \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}
\end{array}$$

Solution:

Two vectors are orthogonal if and only if their dot product is zero.
Therefore, we calculate the dot product for each pair.

Pair 1:

Let

$$u = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Then

$$u \cdot v = (1)(2) + (-1)(0) + (-2)(1).$$

Therefore,

$$u \cdot v = 2 + 0 - 2 = 0.$$

Hence, this pair of vectors is orthogonal.

Pair 2:

$$\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$$

gives

$$(1)(-1) + (-2)(4) + (4)(2) = -1 - 8 + 8 = -1.$$

Since

$$-1 \neq 0,$$

this pair of vectors is not orthogonal.

Pair 3:

$$\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix}$$

gives

$$(1)(-2) + (-2)(5) + (4)(2) = -2 - 10 + 8 = -4.$$

Since

$$-4 \neq 0,$$

this pair of vectors is not orthogonal.

Pair 4:

$$\begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$$

gives

$$(5)(-2) + (-2)(4) + (3)(6).$$

Therefore,

$$-10 - 8 + 18 = 0.$$

Hence, this pair of vectors is also orthogonal.

Therefore, the correct choices are pairs 1 and 4:

$$\boxed{\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}}$$

Both pairs have dot product equal to zero.

Question 10

Suppose the eigen values of matrix A are $-3, -4, -4, -3$, then the determinant of the matrix $(A^{-1})^T$ is

- 0.06
- -14
- 0

✓ 0.0069

Solution:

The eigenvalues of A are

$$-3, \quad -4, \quad -4, \quad -3.$$

The determinant of a matrix is equal to the product of its eigenvalues.

Therefore,

$$\det(A) = (-3)(-4)(-4)(-3).$$

First,

$$(-3)(-4) = 12$$

and

$$(-4)(-3) = 12.$$

Hence,

$$\det(A) = 12 \times 12 = 144.$$

For an invertible matrix,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Therefore,

$$\det(A^{-1}) = \frac{1}{144}.$$

We are asked to find

$$\det((A^{-1})^T).$$

The determinant of a transpose is the same as the determinant of the original matrix:

$$\det(B^T) = \det(B).$$

Therefore,

$$\det((A^{-1})^T) = \det(A^{-1}).$$

Hence,

$$\det((A^{-1})^T) = \frac{1}{144}.$$

Numerically,

$$\frac{1}{144} = 0.006944 \dots$$

which is approximately

$$0.0069.$$

Therefore,

$$\boxed{0.0069}$$

and option (D) is correct.

Week-1

Graded Assignment

Question 1

The function `ls()` in R will

- set a new working directory path
- ✓ list all objects in our working environment
- display the path to our working directory
- list all files in our working directory

Solution:

The function `ls()` is used in R to display the names of objects that are currently present in the working environment.

For example, suppose we create two variables:

```
x <- 10
name <- "Archit"
```

If we execute

```
ls()
```

R displays the objects available in the current environment, such as `x` and `name`.

It is important not to confuse `ls()` with other R functions:

- `setwd()` changes the working directory.
- `getwd()` displays the current working directory.
- `list.files()` lists files in the working directory.
- `ls()` lists objects in the R environment.

Therefore, the correct answer is:

list all objects in our working environment

Hence, option (B) is correct.

Question 2

Consider the following code:

```
room.temperature <- as.integer(
  readline(prompt="Enter room temperature in degree celsius: ")
)
```

```

if (room.temperature < 19) {
    print("The room is chilly. Please turn the heater on!")
} else {
    if (room.temperature >= 19 && room.temperature <= 25) {
        print("The room is warm")
        print("Enjoy the stay!")
    } else {
        print("The room is hot")
        print("Please turn the air conditioning on!")
    }
}

```

If a room temperature of 20 degree Celsius is input to the above code, choose the output displayed.

- "The room is chilly. Please turn the heater on!"
- ✓ "The room is warm" "Enjoy the stay!"
- "The room is hot" "Please turn the air conditioning on!"
- None of the above

Solution:

The input is

$$\text{room.temperature} = 20.$$

The first condition is

$$20 < 19.$$

This condition is **FALSE**, so the first **if** block is not executed.

R therefore moves to the **else** block. Inside it, we check

$$20 \geq 19 \quad \text{and} \quad 20 \leq 25.$$

Both conditions are true:

$$20 \geq 19 = \text{TRUE}$$

and

$$20 \leq 25 = \text{TRUE}.$$

Therefore,

$$\text{TRUE} \ \&\& \ \text{TRUE} = \text{TRUE}.$$

Hence, the inner **if** block is executed:

```

print("The room is warm")

```

```
print("Enjoy the stay!")
```

Thus, the displayed output is:

```
"The room is warm" "Enjoy the stay!"
```

Hence, option (B) is correct.

Question 3

Consider the code below to create a dataframe city_data:

```
name <- c("City A", "City B", "City C")
weather <- c("Sunny", "cloudy", "Rainy")
city_data <- data.frame(name, weather)
```

Choose the correct answer based on the output expected on running the below code.

```
city_data$weather[city_data$name == "City C"] <- "Snowy"
print(city_data)
```

- ✓ In the column weather in dataframe city_data, “Rain” is replaced by “Snowy”
- In the column name in dataframe city_data, “City C” is replaced by “Snowy”
- No change occurs to the entries in the dataframe city_data
- The code raises an error

Solution:

Initially, the dataframe is:

name	weather
City A	Sunny
City B	cloudy
City C	Rainy

The important statement is:

```
city_data$weather[city_data$name == "City C"] <- "Snowy"
```

Let us break this statement into parts.

First,

```
city_data$name == "City C"
```

checks which row has the name "City C".

The resulting logical vector is:

(FALSE, FALSE, TRUE).

Next,

```
city_data$weather[city_data$name == "City C"]
```

selects the weather value corresponding to City C. That value is initially "Rainy".

The assignment

```
<- "Snowy"
```

therefore replaces only that value.

The resulting dataframe is:

name	weather
City A	Sunny
City B	cloudy
City C	Snowy

Therefore, "Rainy" is replaced by "Snowy" in the `weather` column.

Hence, option (A) is correct.

Question 4

Consider the code given below.

```
city_weather = list(  
  "City A",  
  "City B",  
  "City C",  
  c("Sunny", "cloudy", "Rainy")  
)
```

Choose the correct command to access the variable "Cloudy".

- ✓ `city_weather[[4]][2]`
 - `city_weather[[5]]`
 - `city_weather[[2]][2]`
 - None of the above
-

Solution:

The list `city_weather` contains four elements:

Position	Value
1	"City A"
2	"City B"
3	"City C"
4	c("Sunny", "cloudy", "Rainy")

The fourth element is itself a vector:

```
c("Sunny", "cloudy", "Rainy").
```

We use

```
city_weather[[4]]
```

to extract the fourth element of the list.

This gives:

```
c("Sunny", "cloudy", "Rainy").
```

Now we need the second element of this vector. Therefore, we use

```
city_weather[[4]][2]
```

which gives

"cloudy"

.

Thus, the correct command is:

city_weather[[4]][2]

Hence, option (A) is correct.

Question 5

Consider the following code.

```
A = matrix(c(1:42), nrow = 6, ncol = 7, byrow = T)
B = A[-2,]
```

Based on the output of the above code, choose the correct option.

- ☐ B is a matrix consisting of only elements of the second row of matrix A
- ☐ B is a matrix consisting of same elements as that of A
- ☐ The code raises an error
- ☒ B is a matrix consisting of all elements of A excluding the second row

Solution:

The matrix is created using:

```
A = matrix(c(1:42), nrow = 6, ncol = 7, byrow = T)
```

Since there are 42 elements and the matrix has 6 rows and 7 columns, the matrix is:

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 8 & 9 & 10 & 11 & 12 & 13 & 14 \\ 15 & 16 & 17 & 18 & 19 & 20 & 21 \\ 22 & 23 & 24 & 25 & 26 & 27 & 28 \\ 29 & 30 & 31 & 32 & 33 & 34 & 35 \\ 36 & 37 & 38 & 39 & 40 & 41 & 42 \end{bmatrix}.$$

The next statement is:

```
B = A[-2,]
```

In R, a negative index means that the corresponding row or column should be excluded.
Here, -2 means:

exclude row 2.

The comma after -2 indicates that all columns should be retained.
Therefore, B contains rows

1, 3, 4, 5, 6.

Hence,

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 15 & 16 & 17 & 18 & 19 & 20 & 21 \\ 22 & 23 & 24 & 25 & 26 & 27 & 28 \\ 29 & 30 & 31 & 32 & 33 & 34 & 35 \\ 36 & 37 & 38 & 39 & 40 & 41 & 42 \end{bmatrix}.$$

Thus, B consists of all elements of A except the second row.
Therefore, option (D) is correct.

Question 6

The parameter “collapse” in the function `paste()`

- adjusts the display of the string to left, right or center
- add space in between two strings
- ✓ eliminates the space in between two strings
- eliminates the space within two words in a string

Solution:

The `paste()` function is used to combine strings.

The `collapse` parameter specifies the separator that should be used when multiple elements are combined into a single string.

For example,

```
paste(c("Hello", "World"), collapse = " ")
```

produces:

```
"Hello World".
```

Here, the value of `collapse` is a space:

```
collapse = " ".
```

If we use an empty string,

```
paste(c("Hello", "World"), collapse = "")
```

the two elements are joined without a space:

"HelloWorld".

Thus, when `collapse=""` is used, the space/separator between the two strings is eliminated.

Therefore, according to the given question, the correct answer is:

eliminates the space in between two strings

Hence, option (C) is correct.

Question 7

Consider the following code in R.

```
find.function <- function(a) {  
  for(i in 1:a) {  
    b <- i^2  
    print(b)  
  }  
}
```

If we supply 4 as an argument to the function `find.function`, after executing the above code, the result displayed would be

- ☐ the numbers "1,2,3,4"
- ☒ the numbers "1,4,9,16"
- ☐ the number "4"
- ☐ None of the above

Solution:

The function is:

```
find.function <- function(a) {  
  for(i in 1:a) {  
    b <- i^2  
    print(b)  
  }  
}
```

We supply

$a = 4$.

Therefore, the loop

```
for(i in 1:a)
```

becomes

$$i = 1, 2, 3, 4.$$

For every value of i , the statement

```
b <- i^2
```

calculates the square of i .

The values are:

i	i^2
1	1
2	4
3	9
4	16

The `print(b)` statement therefore displays:

1, 4, 9, 16.

Hence, the correct answer is:

the numbers "1,4,9,16"

Therefore, option (B) is correct.

Question 8

The operator `%in%` is used for

- multiplication of a matrix with its transpose
- generating a sequence of numbers in a vector
- ✓ identifying if an element belongs to a vector
- none of the above

Solution:

The operator `%in%` is the membership operator in R.

It checks whether an element is present in a vector.

For example,

```
3 %in% c(1, 2, 3, 4)
```

returns:

TRUE

because 3 is present in the vector.

Similarly,

```
7 %in% c(1, 2, 3, 4)
```

returns:

FALSE

because 7 is not present in the vector.

Therefore,

<code>%in%</code> identifies if an element belongs to a vector
--

Hence, option (C) is correct.

Question 9

The output of compiling and executing the following code in R would be

```
v_name <- c("Green", "apple")
cnt <- 2
```

```
repeat {
  print(v_name)
  cnt <- cnt + 1

  if(cnt > 5) {
    break
  }
}
```

- The term “Green” “apple” printed 5 times.
- The term “Green apple” printed 3 times.
- The term “Green” printed 4 times
- ✓ The term “Green” “apple” printed 4 times.

Solution:

The vector is initialized as:

```
v_name <- c("Green", "apple")
```

Therefore, every execution of

```
print(v_name)
```

prints the two elements:

"Green" and "apple".

The counter starts with

$$cnt = 2.$$

The **repeat** loop first prints the vector and then executes

```
cnt <- cnt + 1
```

After each iteration, the value of **cnt** is:

Iteration	cnt before increment	cnt after increment
1	2	3
2	3	4
3	4	5
4	5	6

After the fourth iteration,

$$cnt = 6.$$

The condition

```
if(cnt > 5)
```

becomes

$$6 > 5,$$

which is **TRUE**. Therefore, **break** terminates the loop.

Hence, **print(v.name)** executes exactly four times.

Therefore,

The term “Green” “apple” is printed 4 times.

Hence, option (D) is correct.

Question 10

The “**next**” statement in R programming is useful

- ✓ for skipping the current iteration of a loop without terminating it
- for terminating the current iteration of a loop
- for evaluating the current iteration of a loop without terminating it
- None of the above

Solution:

The **next** statement is used inside loops.

When R encounters **next**, it skips the remaining statements in the current iteration and

immediately proceeds to the next iteration of the loop.

For example:

```
for(i in 1:5) {  
    if(i == 3) {  
        next  
    }  
    print(i)  
}
```

The loop starts with

$$i = 1.$$

It prints 1.

For $i = 2$, it prints 2.

When

$$i = 3,$$

the condition

```
i == 3
```

is true. Therefore, **next** is executed. The remaining statements in that iteration are skipped, so 3 is not printed.

The loop then continues with $i = 4$ and $i = 5$.

The output is therefore:

1, 2, 4, 5.

It is important to distinguish **next** from **break**:

- **next**: skips the current iteration and continues the loop.
- **break**: terminates the entire loop.

Therefore, the correct answer is:

for skipping the current iteration of a loop without terminating it

Hence, option (A) is correct.

Week 3 : Assignment 3

Detailed Solutions

Question 1

Sumit wants to contact one of his friends, but he remembers only the first 9 of the 10 digits of the contact number. He is sure that the last digit of the contact number is an odd number. He selects an odd number randomly. If the random variable X denotes the last digit of the contact number, calculate $\text{Var}(X)$.

- ☐ 5
- ☒ 8
- ☐ 33
- ☐ None of the above

Solution:

The possible odd digits are

$$X \in \{1, 3, 5, 7, 9\}.$$

Since one of the five odd digits is selected randomly, each value has probability

$$P(X = x) = \frac{1}{5}.$$

Therefore,

$$E(X) = \frac{1 + 3 + 5 + 7 + 9}{5}.$$

Hence,

$$E(X) = \frac{25}{5} = 5.$$

Now calculate $E(X^2)$:

$$E(X^2) = \frac{1^2 + 3^2 + 5^2 + 7^2 + 9^2}{5} = \frac{1 + 9 + 25 + 49 + 81}{5}.$$

$$E(X^2) = \frac{165}{5} = 33.$$

The variance is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 33 - 5^2 = 33 - 25 = 8.$$

$\text{Var}(X) = 8$

Hence, the correct answer is 8.

Question 2

Suppose $X \sim N(\mu, 4)$. For $n = 20$ iid samples of X , the observed sample mean is $\bar{X} = 5.2$. What conclusion would a z -test reach if the null hypothesis assumes $\mu = 5$ against the alternative hypothesis $\mu \neq 5$ at significance level $\alpha = 0.05$? Use $F_Z^{-1}(0.025) = -1.9599$.

- ✓ Accept H_0
 - Reject H_0
-

Solution:

We have

$$X \sim N(\mu, 4).$$

Since the variance is 4, the standard deviation is

$$\sigma = \sqrt{4} = 2.$$

The hypotheses are

$$H_0 : \mu = 5 \quad \text{and} \quad H_1 : \mu \neq 5.$$

Since the alternative hypothesis contains $\neq$, this is a **two-tailed test**.

The z -test statistic is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Here,

$$\bar{X} = 5.2, \quad \mu_0 = 5, \quad \sigma = 2, \quad n = 20.$$

Therefore,

$$Z = \frac{5.2 - 5}{2/\sqrt{20}} = \frac{0.2}{2/\sqrt{20}}.$$

Since $\sqrt{20} \approx 4.4721$, we get $2/\sqrt{20} \approx 0.4472$.

Hence,

$$Z \approx \frac{0.2}{0.4472} \approx 0.4472.$$

For a two-tailed test at $\alpha = 0.05$, the critical values are

$$-1.9599 \quad \text{and} \quad 1.9599.$$

Thus, we reject H_0 only if $Z < -1.9599$ or $Z > 1.9599$. But

$$-1.9599 < 0.4472 < 1.9599.$$

Therefore, the test statistic does not fall in the rejection region.

Accept H_0

The correct option is **Accept H_0** .

Question 3

A box contains 8 items of which 2 are defective. A sample of 5 items is selected randomly without replacement. If the random variable X represents the number of defective items in the sample, find $E(X)$.

- ✓ 1.25
 - 5
-

Solution:

This is a **hypergeometric sampling** situation because:

- There are $N = 8$ total items.
- There are $K = 2$ defective items.
- We select $n = 5$ items.
- Sampling is without replacement.

For a hypergeometric random variable, the expected value is

$$E(X) = n \frac{K}{N}.$$

Substituting the values,

$$E(X) = 5 \left(\frac{2}{8} \right) = 5 \left(\frac{1}{4} \right) = \frac{5}{4} = 1.25.$$

$E(X) = 1.25$

The correct answer is **1.25**.

Question 4

Suppose $X \sim N(\mu, 9)$. For $n = 100$ iid samples of X , the observed sample mean is $\bar{X} = 11.8$. What conclusion would a z -test reach if the null hypothesis assumes $\mu = 10.5$ against an alternative hypothesis $\mu \neq 10.5$? Consider significance levels 0.10 and 0.05. (Select all that apply.)

- ☐ Accept H_0 at a significance level of 0.10.
- ☒ Reject H_0 at a significance level of 0.10.
- ☐ Accept H_0 at a significance level of 0.05.
- ☒ Reject H_0 at a significance level of 0.05.

Solution:

We have

$$X \sim N(\mu, 9) \quad \Rightarrow \quad \sigma = \sqrt{9} = 3.$$

The hypotheses are

$$H_0 : \mu = 10.5 \quad \text{and} \quad H_1 : \mu \neq 10.5.$$

Since the alternative hypothesis is $\mu \neq 10.5$, this is a two-tailed test. The z -statistic is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{11.8 - 10.5}{3/\sqrt{100}} = \frac{1.3}{3/10} = \frac{1.3}{0.3} = 4.3333.$$

So the calculated test statistic is $Z \approx 4.33$.

At significance level $\alpha = 0.10$: For a two-tailed test, $\alpha/2 = 0.05$, giving critical values ± 1.645 . Since $4.33 > 1.645$, the test statistic lies in the rejection region.

Reject H_0 at $\alpha = 0.10$

At significance level $\alpha = 0.05$: For a two-tailed test, $\alpha/2 = 0.025$, giving critical values ± 1.96 . Since $4.33 > 1.96$, the test statistic again lies in the rejection region.

Reject H_0 at $\alpha = 0.05$

The correct choices are **Reject H_0 at significance level 0.10** and **Reject H_0 at significance level 0.05**.

Question 5

Let X and Y be two independent random variables with $\text{Var}(X) = 9$ and $\text{Var}(Y) = 3$. Find $\text{Var}(4X - 2Y + 6)$.

- ☐ 100
- ☐ 140

- ✓ 156
- None of the above

Solution:

We use the variance rule

$$\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Since X and Y are independent, $\text{Cov}(X, Y) = 0$. Therefore,

$$\text{Var}(4X - 2Y + 6) = 4^2 \text{Var}(X) + (-2)^2 \text{Var}(Y).$$

The constant 6 does not affect variance. Thus,

$$= 16(9) + 4(3) = 144 + 12 = 156.$$

$\text{Var}(4X - 2Y + 6) = 156$

The correct option is **156**.

Question 6

The correlation coefficient of two random variables X and Y is 0.14. Their variances are given by 3 and 5. Compute $\text{Cov}(X, Y)$.

- -0.854
- 0.561
- ✓ 0.54
- None of the above

Solution:

The relationship between correlation and covariance is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad \Rightarrow \quad \text{Cov}(X, Y) = \rho_{XY} \sigma_X \sigma_Y.$$

We are given $\rho_{XY} = 0.14$, $\text{Var}(X) = 3$, and $\text{Var}(Y) = 5$, so

$$\sigma_X = \sqrt{3}, \quad \sigma_Y = \sqrt{5}.$$

Hence,

$$\text{Cov}(X, Y) = 0.14(\sqrt{3})(\sqrt{5}) = 0.14\sqrt{15}.$$

Since $\sqrt{15} \approx 3.873$,

$$\text{Cov}(X, Y) \approx 0.14(3.873) \approx 0.5422.$$

Rounding to two decimal places,

$\text{Cov}(X, Y) \approx 0.54$

The correct option is **0.54**.

Question 7

When will you reject the null hypothesis?

- ☐ p value greater than α
- ☒ p value less than α
- ☐ p value equal to α
- ☐ None of the above

Solution:

In hypothesis testing, the p -value is compared with the significance level α . The decision rule is:

Reject H_0 if $p\text{-value} < \alpha$.

If $p\text{-value} > \alpha$, we do not reject H_0 .

The correct option is p **value less than** α .

Question 8

A sample of N observations is independently drawn from a normal distribution. The sample variance follows which distribution?

- ☐ Normal distribution
- ☐ Chi-square with N degrees of freedom
- ☒ Chi-square with $N - 1$ degrees of freedom
- ☐ t -distribution with $N - 1$ degrees of freedom

Solution:

Suppose $X_1, X_2, \dots, X_N$ are independent observations from a normal distribution $N(\mu, \sigma^2)$. The sample variance is

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2.$$

For a random sample from a normal population, the following result holds:

$$\frac{(N-1)S^2}{\sigma^2} \sim \chi_{N-1}^2.$$

Thus, the quantity involving the sample variance follows a chi-square distribution with $N - 1$ degrees of freedom.

Chi-square distribution with $N - 1$ degrees of freedom

The correct option is **Chi-square with $N - 1$ degrees of freedom.**

Question 9

A car manufacturer purchases car batteries from two different suppliers. Supplier X provides 55% of the batteries and Supplier Y provides the rest. 5% of all batteries from Supplier X are defective and 4% of all batteries from Supplier Y are defective. You select a battery from the bulk and find it to be defective. What is the probability that it is from Supplier X ?

- ☐ 0.0455
- ☐ 0.455
- ☒ 0.60
- ☐ 0.018

Solution:

We need to find $P(X | D)$, where X = battery is from Supplier X , Y = battery is from Supplier Y , and D = battery is defective.

We are given

$$P(X) = 0.55 \quad \Rightarrow \quad P(Y) = 1 - 0.55 = 0.45.$$

The probabilities of a defective battery are

$$P(D | X) = 0.05 \quad \text{and} \quad P(D | Y) = 0.04.$$

We use Bayes' theorem:

$$P(X | D) = \frac{P(D | X)P(X)}{P(D)}.$$

First calculate $P(D)$ using the law of total probability:

$$P(D) = P(D | X)P(X) + P(D | Y)P(Y) = (0.05)(0.55) + (0.04)(0.45).$$

Calculate each term: $(0.05)(0.55) = 0.0275$ and $(0.04)(0.45) = 0.018$.

Therefore,

$$P(D) = 0.0275 + 0.018 = 0.0455.$$

Now apply Bayes' theorem:

$$P(X | D) = \frac{(0.05)(0.55)}{0.0455} = \frac{0.0275}{0.0455} \approx 0.6044.$$

Rounding to two decimal places,

$$P(X | D) \approx 0.60$$

Therefore, the probability that the defective battery came from Supplier X is approximately **0.60**.

Question 10

Which one of the following is best measure of central tendency for categorical data?

- Mean
- Median
- ✓ Mode
- None of the above

Solution:

Categorical data consists of values that represent categories or labels (such as colors, brands, or types) rather than numerical quantities that can be meaningfully added, subtracted, or ordered on a numeric scale.

Mean requires numerical values that can be summed and divided, which is not meaningful for categories. **Median** requires the data to be ordered on a numeric or at least ordinal scale, which categorical (nominal) data does not generally support.

Mode, on the other hand, simply identifies the most frequently occurring category. Since it only relies on counting frequencies rather than performing arithmetic or ordering operations, it is well-defined for categorical data.

For example, if a survey records favourite fruit among {Apple, Banana, Apple, Mango, Apple}, the mode is “Apple,” since it occurs most frequently. Neither a mean nor a median can be meaningfully computed for this data.

Mode

The correct option is **Mode**.

Answer Summary

Question	Correct Answer
1	8
2	Accept H_0
3	1.25
4	Reject H_0 at 0.10 and 0.05
5	156
6	0.54
7	$p\text{-value} < \alpha$
8	χ^2 with $N - 1$ degrees of freedom
9	0.60
10	Mode

Week 4 : Assignment 4

Detailed Solutions

Question 1

Let $f(x) = x^3 + 6x^2 - 3x - 5$. Select the correct options from the following:
(Select all that apply.)

- ☐ $-2 + \sqrt{5}$ will give the maximum for $f(x)$.
 - ☒ $-2 + \sqrt{5}$ will give the minimum for $f(x)$.
 - ☒ The stationary points for $f(x)$ are $-2 + \sqrt{5}$ and $-2 - \sqrt{5}$.
 - ☐ The stationary points for $f(x)$ are -4 and 0 .
-

Solution:

A stationary point occurs where the first derivative is zero.

Given

$$f(x) = x^3 + 6x^2 - 3x - 5.$$

Differentiate:

$$f'(x) = 3x^2 + 12x - 3.$$

For stationary points, $f'(x) = 0$, so

$$3x^2 + 12x - 3 = 0.$$

Dividing by 3,

$$x^2 + 4x - 1 = 0.$$

Using the quadratic formula,

$$x = \frac{-4 \pm \sqrt{4^2 - 4(1)(-1)}}{2} = \frac{-4 \pm \sqrt{20}}{2}.$$

Since $\sqrt{20} = 2\sqrt{5}$, we obtain

$$x = -2 \pm \sqrt{5}.$$

Hence the stationary points are

$$\boxed{x = -2 + \sqrt{5}, \quad x = -2 - \sqrt{5}}.$$

Now use the second derivative to classify them.

$$f''(x) = 6x + 12.$$

At $x = -2 + \sqrt{5}$,

$$f''(-2 + \sqrt{5}) = 6(-2 + \sqrt{5}) + 12 = 6\sqrt{5} > 0.$$

Since the second derivative is positive, this point is a local **minimum**.

At $x = -2 - \sqrt{5}$,

$$f''(-2 - \sqrt{5}) = -6\sqrt{5} < 0,$$

so this point gives a local **maximum**.

The correct options are: $-2 + \sqrt{5}$ gives the minimum for $f(x)$, and the stationary points are $-2 + \sqrt{5}$ and $-2 - \sqrt{5}$.

Use the following information for Questions 2 and 3.

Consider the optimization problem

$$\max_{x \in \mathbb{R}} f(x), \quad \text{where} \quad f(x) = x^4 + 7x^3 + 5x^2 - 17x + 3.$$

Let x^* be the maximizer of $f(x)$.

Question 2

What is the second order sufficient condition for x^* to be the maximizer of the function $f(x)$?

- $4x^3 + 21x^2 + 10x - 17 = 0$
- $12x^2 + 42x + 10 = 0$
- $12x^2 + 42x + 10 > 0$
- ✓ $12x^2 + 42x + 10 < 0$

Solution:

For a differentiable function, a necessary condition for an interior stationary point is

$$f'(x^*) = 0.$$

For a maximum, the second derivative must be negative:

$$\boxed{f''(x^*) < 0}.$$

First calculate the derivatives. Given

$$f(x) = x^4 + 7x^3 + 5x^2 - 17x + 3,$$

we get

$$f'(x) = 4x^3 + 21x^2 + 10x - 17.$$

Differentiating again,

$$f''(x) = 12x^2 + 42x + 10.$$

Therefore, the second-order sufficient condition for a maximum is

$$\boxed{12(x^*)^2 + 42x^* + 10 < 0}.$$

Question 3

Find the value of x^* .

- ☐ -4.48
 - ☐ 0.66
 - ☒ -1.43
 - ☐ 4.45
-

Solution:

For a stationary point, $f'(x) = 0$. We already found

$$f'(x) = 4x^3 + 21x^2 + 10x - 17.$$

Therefore, we solve

$$4x^3 + 21x^2 + 10x - 17 = 0.$$

The three roots are approximately

$$x \approx -4.48, \quad x \approx -1.43, \quad x \approx 0.66.$$

To determine which one corresponds to a maximum, use

$$f''(x) = 12x^2 + 42x + 10.$$

For $x \approx -1.43$,

$$f''(-1.43) = 12(-1.43)^2 + 42(-1.43) + 10 \approx 24.53 - 60.06 + 10 \approx -25.53 < 0.$$

Therefore, $x = -1.43$ satisfies the second-order sufficient condition for a maximum.

$$\boxed{x^* \approx -1.43}$$

Question 4

Let $f(x) = 2 \sin x$, $0 \leq x \leq 2\pi$. Select the correct options from the following:
(Select all that apply.)

- ☒ $\frac{\pi}{2}$ is the global maximum of $f(x)$.
- ☐ π is the global minimum of $f(x)$.
- ☐ $\frac{3\pi}{2}$ is the global maximum of $f(x)$.
- ☒ $\frac{3\pi}{2}$ is the global minimum of $f(x)$.

Solution:

Given $f(x) = 2 \sin x$. We know that $-1 \leq \sin x \leq 1$. Multiplying throughout by 2,

$$-2 \leq 2 \sin x \leq 2.$$

Therefore, the maximum value is $f(x) = 2$ when $\sin x = 1$. In the interval $[0, 2\pi]$,

$$\sin x = 1 \quad \text{at} \quad x = \frac{\pi}{2}.$$

Hence, $x = \frac{\pi}{2}$ is the location of the global maximum.

Similarly, the minimum occurs when $\sin x = -1$. In $[0, 2\pi]$,

$$\sin x = -1 \quad \text{at} \quad x = \frac{3\pi}{2}.$$

Hence, $x = \frac{3\pi}{2}$ is the location of the global minimum.

The correct statements are: $\frac{\pi}{2}$ is the global maximum of $f(x)$, and $\frac{3\pi}{2}$ is the global minimum of $f(x)$.

Use the following information for Questions 5, 6, 7 and 8.

Let

$$f(x) = 2x_1^2 + 3x_1x_2 + 3x_2^2 + x_1 + 3x_2.$$

Question 5

Find the gradient for $f(x)$.

- ✓ $\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}$
- $\nabla f = \begin{bmatrix} 3x_1 + 6x_2 + 3 \\ 4x_1 + 3x_2 + 1 \end{bmatrix}$

- $\nabla f = \begin{bmatrix} 4x_1 + 3x_2 \\ 3x_1 + 6x_2 \end{bmatrix}$
- $\nabla f = \begin{bmatrix} 4x_2 + 3x_1 + 1 \\ 3x_2 + 6x_1 + 3 \end{bmatrix}$

Solution:

The gradient is the vector of first-order partial derivatives:

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}.$$

First differentiate with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 4x_1 + 3x_2 + 1.$$

Now differentiate with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = 3x_1 + 6x_2 + 3.$$

Therefore,

$$\boxed{\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}}$$

Question 6

Find the stationary point for $f(x)$.

- 0.6, 0.4
- -0.6, -0.4
- ✓ 0.2, -0.6
- 0.2, 0.6

Solution:

A stationary point occurs when the gradient is zero. From Question 5,

$$\nabla f = \begin{bmatrix} 4x_1 + 3x_2 + 1 \\ 3x_1 + 6x_2 + 3 \end{bmatrix}.$$

Set the gradient equal to zero:

$$4x_1 + 3x_2 = -1 \tag{1}$$

and

$$3x_1 + 6x_2 = -3. \quad (2)$$

From equation (2),

$$x_1 + 2x_2 = -1 \quad \Rightarrow \quad x_1 = -1 - 2x_2.$$

Substitute this into equation (1):

$$4(-1 - 2x_2) + 3x_2 = -1 \quad \Rightarrow \quad -4 - 8x_2 + 3x_2 = -1.$$

Hence,

$$-5x_2 = 3 \quad \Rightarrow \quad x_2 = -\frac{3}{5} = -0.6.$$

Now,

$$x_1 = -1 - 2(-0.6) = -1 + 1.2 = 0.2.$$

Therefore, the stationary point is

$$\boxed{(0.2, -0.6)}.$$

Question 7

Find the Hessian matrix for $f(x)$.

- $\nabla^2 f = \begin{bmatrix} 2 & 3 \\ 3 & 6 \end{bmatrix}$
 - $\nabla^2 f = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$
 - ✓ $\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$
 - $\nabla^2 f = \begin{bmatrix} 6 & 3 \\ 3 & 4 \end{bmatrix}$
-

Solution:

The Hessian matrix contains all second-order partial derivatives:

$$\nabla^2 f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}.$$

From the gradient,

$$\frac{\partial f}{\partial x_1} = 4x_1 + 3x_2 + 1 \quad \Rightarrow \quad \frac{\partial^2 f}{\partial x_1^2} = 4, \quad \frac{\partial^2 f}{\partial x_1 \partial x_2} = 3.$$

Similarly,

$$\frac{\partial f}{\partial x_2} = 3x_1 + 6x_2 + 3 \quad \Rightarrow \quad \frac{\partial^2 f}{\partial x_2 \partial x_1} = 3, \quad \frac{\partial^2 f}{\partial x_2^2} = 6.$$

Thus,

$$\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$$

Question 8

The stationary point obtained in Question 6 is a

- ☐ maxima
 - ☒ minima
 - ☐ saddle point
-

Solution:

From Question 7, the Hessian matrix is

$$H = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}.$$

For a 2×2 Hessian matrix $H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$, it is positive definite if $a > 0$ and $\det(H) > 0$.

Here, $a = 4 > 0$. Also,

$$\det(H) = (4)(6) - (3)(3) = 24 - 9 = 15 > 0.$$

Therefore, the Hessian is positive definite. A stationary point with a positive definite Hessian is a local minimum.

Hence, the stationary point $(0.2, -0.6)$ is a **minimum**.

Question 9

Let $f(x_1, x_2) = 4x_1^2 - 4x_1x_2 + 2x_2^2$. Select the correct options from the following:
(Select all that apply.)

- ☐ $(2, 4)$ is a stationary point of $f(x)$.

- ☒ $(0, 0)$ is a stationary point of $f(x)$.
☒ The Hessian matrix $\nabla^2 f$ is positive definite.
☐ The Hessian matrix $\nabla^2 f$ is not positive definite.
-

Solution:

We first find the gradient. Differentiate with respect to x_1 :

$$\frac{\partial f}{\partial x_1} = 8x_1 - 4x_2.$$

Differentiate with respect to x_2 :

$$\frac{\partial f}{\partial x_2} = -4x_1 + 4x_2.$$

Thus,

$$\nabla f = \begin{bmatrix} 8x_1 - 4x_2 \\ -4x_1 + 4x_2 \end{bmatrix}.$$

For a stationary point, $\nabla f = 0$. Therefore,

$$8x_1 - 4x_2 = 0 \quad \text{and} \quad -4x_1 + 4x_2 = 0.$$

From the first equation, $x_2 = 2x_1$. From the second equation, $x_2 = x_1$. Therefore,

$$x_1 = x_2 = 0.$$

Hence, $(0, 0)$ is the stationary point.

Now calculate the Hessian:

$$\nabla^2 f = \begin{bmatrix} 8 & -4 \\ -4 & 4 \end{bmatrix}.$$

The first leading principal minor is $8 > 0$. The determinant is

$$\det(H) = (8)(4) - (-4)(-4) = 32 - 16 = 16 > 0.$$

Therefore, the Hessian is positive definite. Hence, $(0, 0)$ is a local minimum.

The correct options are: $(0, 0)$ is a stationary point of $f(x)$, and $\nabla^2 f$ is positive definite.

Question 10

In an optimization problem, the function that we want to optimize is called

- Decision function
- Constraints function

- Optimal function
- ✓ Objective function

Solution:

An optimization problem generally has a function whose value we want to maximize or minimize. This function is called the **objective function**.

For example, $\max_x f(x)$ has $f(x)$ as its objective function. Similarly, $\min_x f(x)$ also has $f(x)$ as its objective function.

Objective function

Question 11

The optimization problem $\min_x f(x)$ can also be written as $\max_x f(x)$. Is this statement True or False?

- True
- ✓ False

Solution:

The statement claims

$$\min_x f(x) = \max_x f(x).$$

In general, this is not true. Minimizing a function is equivalent to maximizing its negative:

$$\min_x f(x) = -\max_x [-f(x)].$$

It is **not** generally equivalent to $\max_x f(x)$.

For example, consider $f(x) = x^2$. The minimum occurs at $x = 0$ with value $f(0) = 0$. But x^2 has no finite global maximum over $\mathbb{R}$.

Therefore, $\min_x f(x)$ and $\max_x f(x)$ are clearly not equivalent.

False

Question 12

In the gradient descent algorithm, the step size should always be the same for each iteration. Is this statement True or False?

- True

✓ False

Solution:

The gradient descent update rule is

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k),$$

where α_k is the step size or learning rate at iteration k . The important point is that α_k does not necessarily have to be the same for every iteration.

For example, we can use a constant step size, $\alpha_k = \alpha$. However, we can also use a varying step size, such as $\alpha_k = \frac{1}{k+1}$, which decreases as the number of iterations increases. This is known as a decreasing learning-rate strategy.

Therefore, the step size does **not** have to always remain the same.

False

Answer Summary

Q	Correct Answer
1	$-2 + \sqrt{5}$ gives minimum; stationary points $-2 + \sqrt{5}$, $-2 - \sqrt{5}$
2	$12x^2 + 42x + 10 < 0$
3	$x^* = -1.43$
4	$\pi/2$ global max; $3\pi/2$ global min
5	$\nabla f = [4x_1 + 3x_2 + 1, 3x_1 + 6x_2 + 3]^T$
6	$(0.2, -0.6)$
7	$\nabla^2 f = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix}$
8	Minima
9	$(0, 0)$ is stationary; Hessian positive definite
10	Objective function
11	False
12	False
